

1. Project summary

NativeCAM is a GPL conversational CAM GUI for LinuxCNC (Python 3, GTK3). Maintainer: CNC Proton. Repository: github.com/cnc-proton/nativecam-py3-gtk3

Phase 1 goal: Stabilise the LinuxCNC baseline with automated validation, lathe XZ profile regression, Side Drill checks, and release documentation.

Request: EUR 6,000 / 4 weeks / 130 hours at EUR 45/h

Applicant: Individual, Lithuania (EU)

2. Phase 1 budget and milestones

Work package	Hours	EUR
Lathe XZ stabilization	35	1,575
Side Drill validation	20	900
4th-axis example sanity	15	675
CI / deb / pytest	22	990
VALIDATION.md + release 2.0b-6	12	540
Admin + reporting	8	360
Contingency	18	810
Total	130	6,000

Milestone	EUR	Deliverable
M1 (30%)	1,800	CI green: Actions, pytest, .deb build
M2 (40%)	2,400	Lathe XZ + Side Drill validated with tests
M3 (30%)	1,800	VALIDATION.md, release 2.0b-6, 4th-axis sanity

3. Four-phase roadmap (future - not Phase 1 contract)

Phase	Amount	Weeks	Scope	Submit
1 (this)	EUR 6,000	4	Stabilize baseline, CI, docs	Nov 2026
2	EUR 13,000	8	Operation IR + dual post + 4th axis	May 2027
3	EUR 20,000	10	Siemens ISO subset, dual-output	Nov 2027
4	EUR 26,000	12	5-axis indexing, i18n, packaging	May 2028
Total	EUR ~65,000	34	Multi-controller open CAM	-

Each phase is a separate NLnet proposal after public deliverables of the previous phase.

4. Architecture (future phases)

Feature configs (.cfg)	--> Operation IR (Ph.2)	--> Post-processors
lathe/xz_profile.cfg	JSON/dataclasses	LinuxCncPost
mill/drill-side.cfg		IsoGenericPost (Ph.2)
		SiemensIsoPost (Ph.3)

Phase 1 validates existing Feature -> NGC path only.

5. Prior work and links

- Lathe XZ profiles: G71/G72/G73 on devel - cfg/lathe/xz_profile.cfg, examples/lathe/xz_profile_demo.xml
- Side Drill: cfg/mill/drill-side.cfg (horizontal spindle drilling)
- Validation harness: PR #6 - scripts/validate_project.py, .github/workflows/validate.yml, 40+ pytest tests
- 4th axis demo: examples/mill/4th-axis.xml
- License: GPL. Platform: Debian 13, LinuxCNC 2.9+